# Assignment no 1

## Question no 1:

### Presidents' Ages at First Inauguration

Given data:

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 57 | 61 | 57 | 57 | 58 | 57 | 61 | 54 | 68 |
| 51 | 49 | 64 | 50 | 48 | 65 | 52 | 56 | 46 |
| 54 | 49 | 51 | 47 | 55 | 55 | 54 | 42 | 51 |
| 56 | 55 | 51 | 54 | 51 | 60 | 62 | 43 | 55 |
| 56 | 61 | 52 | 69 | 64 | 46 | 54 | 47 | |

(a) Population or Sample?

These ages include all U.S. Presidents, not just a subset. Therefore, it represents a population, not a sample.

(b) Frequency Distribution:

- Finding max and min value:

  max = 69
  min = 42

- Range:

  Range = max − min
  = 69 − 42
  = 27

- Select number of classes:

  A good choice is 6 classes (common for around 50 data values).

CamScanner

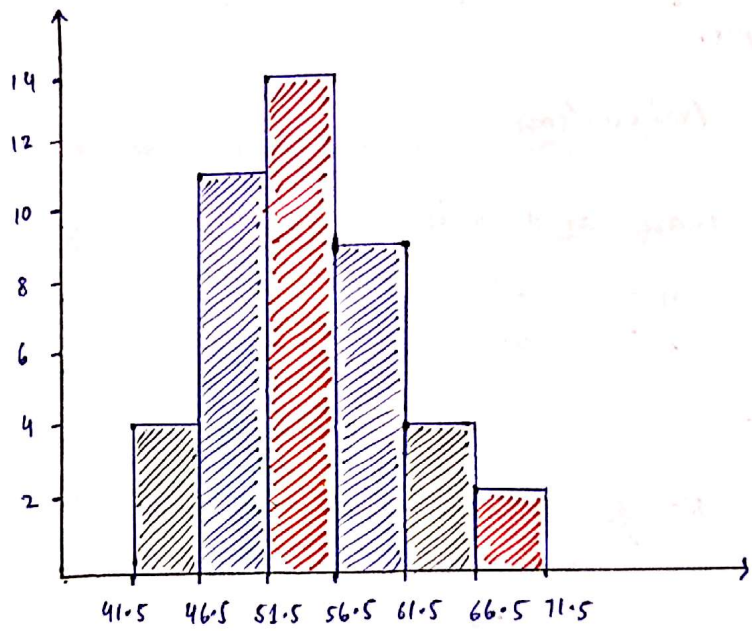
- class width:

$$\text{Width} = \frac{27}{6} = 4.5 \rightarrow \text{round up to } 5$$

- Table:

| class limits | c.b | tally | f | x | c.f |
|---|---|---|---|---|---|
| 42 - 46 | 41.5 - 46.5 | IIII | 4 | 44 | 4 |
| 47 - 51 | 46.5 - 51.5 | ~~IIII~~ ~~IIII~~ I | 11 | 49 | 4+11 = 15 |
| 52 - 56 | 51.5 - 56.5 | ~~IIII~~ ~~IIII~~ IIII | 14 | 54 | 15+14 = 29 |
| 57 - 61 | 56.5 - 61.5 | ~~IIII~~ IIII | 9 | 59 | 29+9 = 38 |
| 62 - 66 | 61.5 - 66.5 | IIII | 4 | 64 | 38+4 = 42 |
| 67 - 71 | 66.5 - 71.5 | II | 2 | 69 | 42+2 = 44 |

- Histogram:




- peaks in distribution:

The histogram shows one main peak, centered near 52-56 age range.

Question no 2:

## Stem and leaf plot

Given Data:

| | | | | | | |
|---|---|---|---|---|---|---|
| 57 | 54 | 52 | 55 | 51 | 56 | 47 |
| 61 | 68 | 56 | 55 | 54 | 61 | 51 |
| 57 | 51 | 46 | 54 | 51 | 52 | |
| 57 | 49 | 54 | 42 | 60 | 69 | |
| 58 | 64 | 49 | 51 | 62 | 64 | |
| 57 | 48 | 51 | 56 | 43 | 46 | |
| 61 | 65 | 47 | 55 | 55 | 54 | |

Arranging data in order:

42, 43, 46, 47, 48, 49, 46, 49, 47, 51, 51, 51, 51, 51, 51, 52, 52, 54, 54, 54, 54, 54, 55, 55, 55, 55, 56, 56, 56, 57, 57, 57, 57, 58, 60, 61, 61, 61, 62, 64, 64, 65, 68, 69

| Stem | Leaf |
|---|---|
| 4 | 2 3 6 6 7 7 8 9 9 |
| 5 | 1 1 1 1 1 1 2 2 4 4 4 4 5 5 5 5 6 6 6 7 7 7 8 |
| 6 | 0 1 1 1 2 4 4 5 8 9 |

Analysis:

- Most presidents took office in their 50s and early 60s
- Distribution is slightly right-skewed (few younger ones in 40s)

Question no 3 :

Commission Data

a) mean

| Class limits | f | Midpoint Xm | f.Xm |
|---|---|---|---|
| 150 - 158 | 5 | 154 | 770 |
| 159 - 167 | 16 | 163 | 2608 |
| 168 - 176 | 20 | 172 | 3440 |
| 177 - 185 | 21 | 181 | 3801 |
| 186 - 194 | 20 | 190 | 2850 |
| 195 - 203 | 15 | 199 | 3980 |
| 204 - 212 | 3 | 208 | 624 |

$\Sigma f = 100$

$\Sigma f \cdot X_m = 18{,}073$

$$\bar{X} = \frac{\Sigma f \cdot X_m}{\Sigma f} = \frac{18073}{100} = 180.73$$

b) Median

| Class | f | c.f |
|---|---|---|
| 150-158 | 5 | 5 |
| 159 - 167 | 16 | 21 |
| 168 - 176 | 20 | 41 |
| 177 - 185 | 21 | 62 |
| 186 - 194 | 20 | 77 |
| 195 - 203 | 15 | 97 |
| 204 - 212 | 3 | 100 |

CamScanner

Median class $= \frac{n}{2} = \frac{100}{2} = 50$ (the one containing 50th value) $\rightarrow$ 177 – 185

Formula:

$$\text{Median} = L + \frac{(n/2 - cf_{prev})}{f} \times h$$

Here,

$L = 176.5$

$n = 100$

$cf_{prev} = 41$

$f = 21$

$h = 9$

$$\text{Median} = 176.5 + \frac{(50 - 41)}{21} \times 9$$

$$= 176.5 + 3.86$$

$$= 180.36$$

Median = 180.36 dollars

Question no 4:

City Temperatures

Given Data:

62, 72, 66, 79, 83, 61, 62, 85, 72, 64, 74, 71,
42, 38, 91, 66, 77, 90, 74, 63, 64, 68, 42

a) Mean, Median, mode:

$$\text{Mean} = \frac{\Sigma x}{n} = \frac{1501}{23} \approx 65.26\,^{\circ}F$$

Median = $12^{th}$ Value = 68°F

Mode = 62, 64, 66, 72, 74 (multi-modal)

b) Best Measure of Central Tendency

=> Median best represents the central tendency because it's not affected by extreme values.

c) Outliers (Using IQR)

$Q1 = 62$

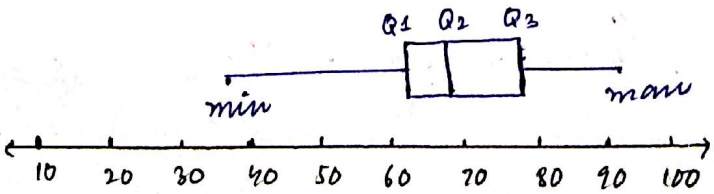
$Q3 = 77$

$IQR = 77 - 62 = 15$

Lower = $62 - 1.5(15) = 39.5$

Upper = $77 + 1.5(15) = 99.5$

Outlier = 38

d) Box and Whisker Plot:

- Min = 38
- Q1 = 62
- median = 68
- Q3 = 77
- max = 91